Breaking 1-WL Indistinguishability on Heterophilous Graphs via Differentiable Class-Separation Structure Learning

Anonymous Author(s)
Under Review at NeurIPS 2023
anonymous@example.com

Abstract

Message-passing graph neural networks (MPGNNs) are theoretically equivalent to the 1-Weisfeiler-Lehman (1-WL) graph isomorphism test, a limitation that is especially severe on heterophilous graphs where neighboring nodes tend to belong to different classes. On such graphs, the original adjacency matrix $\mathbf{A}$ causes message passing to aggregate predominantly cross-class signals—a phenomenon we term aggregation pollution—and the 1-WL coloring fails to distinguish many cross-class node pairs. We propose AGSN (Adaptive Graph Structure Network), a framework that replaces the fixed adjacency with a differentiably learned structure $\mathbf{A}'$ satisfying an (ε, δ)-class-separation condition. We prove two complementary results: an Existence Lemma showing that heterophilous graphs with inter-class edge fraction $\approx 78\%$ cause non-trivial aggregation pollution, and a Recovery Lemma showing that any $\mathbf{A}'$ with class-separation ratio CSR > 1 breaks 1-WL cross-class indistinguishability in a single message-passing step. Empirically, AGSN satisfies CSR > 1 in all 15 runs across Chameleon, Squirrel, and Actor (Platonov 2023 splits), reducing inter-class edge fractions by 11.9–18.1 percentage points. On Actor, AGSN achieves $36.21 \pm 1.08\%$, outperforming all six fixed-structure baselines. An explicit class-separation loss $\mathcal{L}_{\text{sep}}$ is shown to be necessary: removing it collapses structure learning entirely, yielding an accuracy drop of 9.17 percentage points on Chameleon.

1 Introduction

Graph neural networks (GNNs) have become the dominant paradigm for learning on graph-structured data, achieving strong results in tasks ranging from social network analysis to molecular property prediction [8, 15, 18]. Yet their theoretical limitations are well understood: any message-passing GNN (MPGNN) is at most as expressive as the 1-Weisfeiler-Lehman (1-WL) graph isomorphism test [12, 18].

This limitation is acute on heterophilous graphs, where edges tend to connect nodes of different classes. Benchmark datasets such as Chameleon, Squirrel, and Actor (Pei et al. (year?); revised by Platonov et al. (year?)) exhibit inter-class edge fractions of 76–79%. When an MPGNN aggregates a neighborhood dominated by nodes from other classes, the aggregated representation is polluted by irrelevant class signals—a phenomenon we term aggregation pollution. This prevents the 1-WL color assignment from separating cross-class node pairs, even when node features are moderately discriminative.

Existing responses to heterophily fall into two families. The first adapts the aggregation function: multi-hop neighborhoods (H2GCN [19]), spectral polynomial filters (GPRGNN [4], BernNet [6], ACM-GNN [10]), or frequency-adaptive methods (FAGCN [2]). These methods operate on the fixed, original adjacency; they can mitigate but cannot eliminate the structural constraint imposed by a heavily heterophilous graph.

The second family, graph structure learning (GSL), replaces or refines the adjacency during training [3, 5, 7, 20]. However, existing GSL methods are largely designed for homophilous graphs or do not provide a theoretical account of how a learned structure relates to WL expressivity.

Our contribution. We provide, to our knowledge, the first joint theoretical and empirical treatment of structure learning specifically for 1-WL recovery on heterophilous graphs:

1. Existence Lemma (§3): On graphs with inter-class edge fraction $\rho \approx 0.78$ and $K = 5$ balanced classes, a non-trivial fraction of cross-class node pairs are 1-WL indistinguishable because their neighborhood distributions are polluted by the same cross-class mixture.

2. Recovery Lemma (§3): Any learned $\mathbf{A}'$ satisfying the (ε, δ)-class-separation condition (equivalently CSR > 1) breaks this indistinguishability in exactly one 1-WL iteration.

3. AGSN architecture (§4): A memory-efficient factored structure learner ($\mathrm{O}(N \cdot d')$ vs. $\mathrm{O}(N^2 d)$ for

naïve pairwise MLP), an explicit class-separation loss $\mathcal{L}_{\text{sep}}$, and a GIN backbone, all derived from the Recovery Lemma's requirements.

4. Empirical validation (§5): CSR > 1 in all 15 runs; inter-class edge fractions reduced by 11.9–18.1 pp; best accuracy on Actor (36.21%); competitive on Chameleon (40.70%); honest failure analysis on Squirrel.

We use the Platonov et al. (year?) de-duplicated splits throughout, which correct for the data contamination issues identified in the original Pei et al. 2020 benchmark.

2 Related Work

GNN expressivity. The equivalence between MPGNNs and 1-WL was established by Xu et al. (year?) and Morris et al. (year?). Higher-order k-WL tests achieve greater discrimination at $O(n^k)$ cost [1]. Our work exploits the insight that, rather than escalating to k-WL, one can change the graph to make 1-WL sufficient for the classification task at hand.

Heterophily-aware GNNs. H2GCN [19] separates ego and neighbor representations and aggregates multi-hop neighborhoods. GPRGNN [4] learns an arbitrary polynomial filter over graph eigenvalues. BernNet [6] parameterizes filters via Bernstein polynomials. ACM-GNN [10] applies adaptive channel mixing. FAGCN [2] uses frequency-adaptive coefficients. All operate on the original adjacency; they are architecturally limited by its structural properties.

Graph structure learning. LDS [5] casts structure learning as bilevel optimization. IDGL [3] iteratively refines graph and representations. SLAPS [20] uses self-supervised signals. ProGNN [7] targets robust structure recovery. SUBLIME [11] uses contrastive learning. GloGNN [9] learns a global node compatibility matrix—the closest architecture to AGSN—but without theoretical analysis. NodeFormer [16] and SGFormer [17] achieve implicit structure via efficient attention. None of these methods provide a WL-based expressivity analysis specific to heterophilous graphs or include a principled class-separation training objective.

Benchmark contamination. Platonov et al. (year?) demonstrate that the original Chameleon and Squirrel datasets (Pei et al. (year?)) contain duplicate nodes and split leakage, inflating reported baselines by up to 20 percentage points. We use only the corrected, de-duplicated versions.

3 Theoretical Framework

3.1 Preliminaries

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an undirected graph with $n = |\mathcal{V}|$ nodes, adjacency matrix $\mathbf{A} \in \{0,1\}^{n\times n}$, and node feature matrix $\mathbf{X} \in \mathbb{R}^{n\times d}$. Each node $v \in \mathcal{V}$ has a class label $y_v \in \{1, \ldots, K\}$. The inter-class edge fraction is $\rho(\mathbf{A}) = |\{(u,v) \in \mathcal{E} : y_u \neq y_v\}|/|\mathcal{E}|$.

Definition 1 (1-WL Coloring). The 1-WL coloring assigns each node v at iteration t the color $c_v^{(t)} = \text{hash}(c_v^{(t-1)}, \{\{c_u^{(t-1)} : u \in \mathcal{N}(v)\}\})$, where $\mathcal{N}(v)$ denotes the neighborhood of v under $\mathbf{A}$. Two nodes are 1-WL indistinguishable if they receive identical colors at all iterations.

Definition 2 (Aggregation Pollution). For a node v with label y_v, the aggregation pollution ratio is $\pi_v = |\{u \in \mathcal{N}(v) : y_u \neq y_v\}|/\deg(v)$. The graph-level pollution is $\bar{\pi}(\mathbf{A}) = \mathbb{E}_v[\pi_v]$. On a balanced K-class graph with $\rho \approx 0.78$, we have $\bar{\pi} \approx 0.78 \cdot K/(K-1) \approx 0.975$ for $K = 5$.

3.2 Existence Lemma: Heterophilous Graphs Cause WL Indistinguishability

Lemma 1 (Existence Lemma). Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph with K balanced classes, inter-class edge fraction $\rho > (K-1)/K$, and let two nodes u, v have the same degree d and identical degree sequences in their neighborhoods. Then, under topology-only 1-WL coloring (i.e., initial colors $c_u^{(0)} = c_v^{(0)}$), u and v are 1-WL indistinguishable at iteration 1 regardless of their labels $y_u \neq y_v$.

Furthermore, on graphs with $\rho \approx 0.78$ and $K = 5$ (as in Chameleon, Squirrel, Actor), the fraction of cross-class node pairs subject to this collision is measurable and positive (empirically 3–10% under topology-only coloring; see Table 5).

Proof. Fix any two nodes $u, v \in \mathcal{V}$ with $y_u \neq y_v$, equal degree d, and identical local topology (same neighborhood degree sequence). Under topology-only 1-WL coloring, let $c_w^{(0)} = c_0$ for all w. At iteration 1:

$$c_v^{(1)} = \text{hash}(c_0, \{\{c_0, \ldots, c_0\}\}_d) = c_1$$

for any node of degree d, regardless of its class label. Therefore $c_u^{(1)} = c_v^{(1)}$, establishing 1-WL indistinguishability.

The key condition $\rho > (K-1)/K$ ensures that this scenario is not vacuous: if most edges are cross-class, then many nodes of equal degree share the same neighborhood degree sequence purely due to the uniform mixing of classes. On balanced K-class graphs with $\rho > (K-1)/K$ and n/K nodes per class, the fraction of cross-class pairs with equal degree is lower-bounded by a function of the

degree distribution; empirically we measure 3–10% (Table 5).

Even with high-dimensional continuous features (which trivially distinguish nodes in topology-only WL), the aggregation pollution (Definition 2) remains: the GNN's aggregated representation $h_v^{(1)} = \mathrm{AGG}(\{h_u^{(0)} : u \in \mathcal{N}(v)\})$ is dominated by $\approx 78\%$ cross-class features, corrupting the class-relevant signal regardless of how informative individual node features are. □

3.3 Class-Separation Condition and Recovery Lemma

Definition 3 ((ε, δ)-Class-Separation). An adjacency matrix $\mathbf{A}'$ is (ε, δ)-class-separating if for every node v, the intra-class edge weight fraction satisfies $\varepsilon_v = \sum_{u:y_u=y_v} A'_{vu} / \sum_u A'_{vu} \geq \varepsilon$, while for nodes $v' \neq v$ of a different class, $\varepsilon_{v'} \leq \delta$, with $\varepsilon - \delta > 1/K$. Equivalently, the class-separation ratio $\mathrm{CSR}(\mathbf{A}') = \bar{w}_{\text{intra}}/\bar{w}_{\text{inter}} > 1$ where $\bar{w}_{\text{intra}}$ (resp. $\bar{w}_{\text{inter}}$) is the mean edge weight among intra-class (resp. inter-class) node pairs under $\mathbf{A}'$.

Lemma 2 (Recovery Lemma). If $\mathbf{A}'$ is (ε, δ)-class-separating with $\varepsilon - \delta > 1/K$, then for any cross-class pair (u, v) that is 1-WL indistinguishable under $\mathbf{A}$, the 1-WL multiset aggregation at hop 1 under $\mathbf{A}'$ yields distinct colors for u and v. That is, 1-WL indistinguishability is broken in exactly one iteration.

Proof. Under the (ε, δ)-class-separation condition, for node u with label y_u, the normalized intra-class edge weight is at least ε:

$$\sum_{w:y_w=y_u} A'_{uw} \Big/ \sum_w A'_{uw} \;\geq\; \varepsilon.$$

For node v with $y_v \neq y_u$, the intra-class weight for class y_u is at most δ (by definition of the separating condition). Consequently, the 1-WL multiset at hop 1 under $\mathbf{A}'$ gives:

$$M_u = \{\{h_w^{(0)} \cdot A'_{uw} : w \in \mathcal{N}^{\mathbf{A}'}(u)\}\}, \quad M_v = \{\{h_w^{(0)} \cdot A'_{vw} : w \in \mathcal{N}^{\mathbf{A}'}(v)\}\}.$$

The class-y_u weight mass in M_u exceeds that in M_v by at least $\varepsilon - \delta > 1/K > 0$. Since any injective hash function distinguishes multisets with different support weights by at least $1/K$ in the dominant class dimension, $\mathrm{hash}(M_u) \neq \mathrm{hash}(M_v)$, so $c_u^{(1)} \neq c_v^{(1)}$. This holds regardless of what colors u and v had under $\mathbf{A}$, completing the proof that 1-WL recovery occurs in exactly one iteration. □

Corollary 3. Any MPGNN with 1-WL power (e.g., GIN) applied to $\mathbf{A}'$ satisfying CSR > 1 on a heterophilous graph with $\rho > (K-1)/K$ can distinguish all cross-class node pairs that were previously 1-WL indistinguishable under $\mathbf{A}$, in exactly one message-passing layer.

3.4 Architectural Consequence

Lemma 2 implies that any structure learner achieving CSR > 1 is sufficient for 1-WL recovery. Crucially, this condition requires a task-supervised gradient signal toward class-discriminative structure. Pure feature-similarity objectives (as used in IDGL, SLAPS) optimize feature reconstruction rather than class boundary placement and do not guarantee CSR > 1 on heterophilous graphs where features and labels are weakly correlated.

4 Method: Adaptive Graph Structure Network

4.1 Architecture Overview

AGSN comprises three components: a factored structure learner, a top-k sparsification step, and a GIN backbone. All components are jointly optimized end-to-end.

Factored structure learner. Naïve pairwise MLP structure learning [3] computes scores as $S_{uv} = f_\theta([\mathbf{X}_u; \mathbf{X}_v])$, requiring $O(N^2 \cdot d)$ memory—infeasible at $N = 890, d = 2325$ (14.8 GB). Instead, we project each node to a low-dimensional embedding:

$$\mathbf{Z}_v = \sigma(W_2\, \sigma(W_1 \mathbf{X}_v + b_1) + b_2), \quad \mathbf{Z}_v \in \mathbb{R}^{d'},$$

and compute structure scores via cosine similarity:

$$S_{uv} = \frac{\mathbf{Z}_u^\top \mathbf{Z}_v}{\|\mathbf{Z}_u\| \cdot \|\mathbf{Z}_v\|}.$$

Memory is $O(N \cdot d' + N^2)$, a $\sim$1000$\times$ reduction (3 MB vs. 14.8 GB for Chameleon). We use $d' = 64$.

Top-k sparsification. To obtain a sparse $\mathbf{A}'$, we retain only the top-k scored edges per node ($k = 20$ by default), applying a ReLU activation to zero negative scores. The resulting adjacency is symmetrized: $A'_{uv} = \max(S_{uv}, S_{vu}) \cdot \mathbf{1}[\text{top-}k]$.

GIN backbone. We use 2-layer GIN [18] as the MPGNN backbone, since GIN achieves the maximum 1-WL expressivity among MPGNNs:

$$H_v^{(\ell+1)} = \mathrm{MLP}^{(\ell)}\left((1+\epsilon) H_v^{(\ell)} + \sum_{u \in \mathcal{N}^{\mathbf{A}'}(v)} A'_{uv} H_u^{(\ell)}\right).$$

4.2 Training Objective

The full loss is

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{sparse}} \mathcal{L}_{\text{sparse}} + \lambda_{\text{sep}} \mathcal{L}_{\text{sep}}, \tag{1}$$

where $\mathcal{L}_{\text{task}}$ is cross-entropy on training node labels, $\mathcal{L}_{\text{sparse}} = \|S\|_1 / N^2$ penalizes dense structure, and

$$\mathcal{L}_{\text{sep}} = \bar{w}_{\text{inter}} - \bar{w}_{\text{intra}}$$

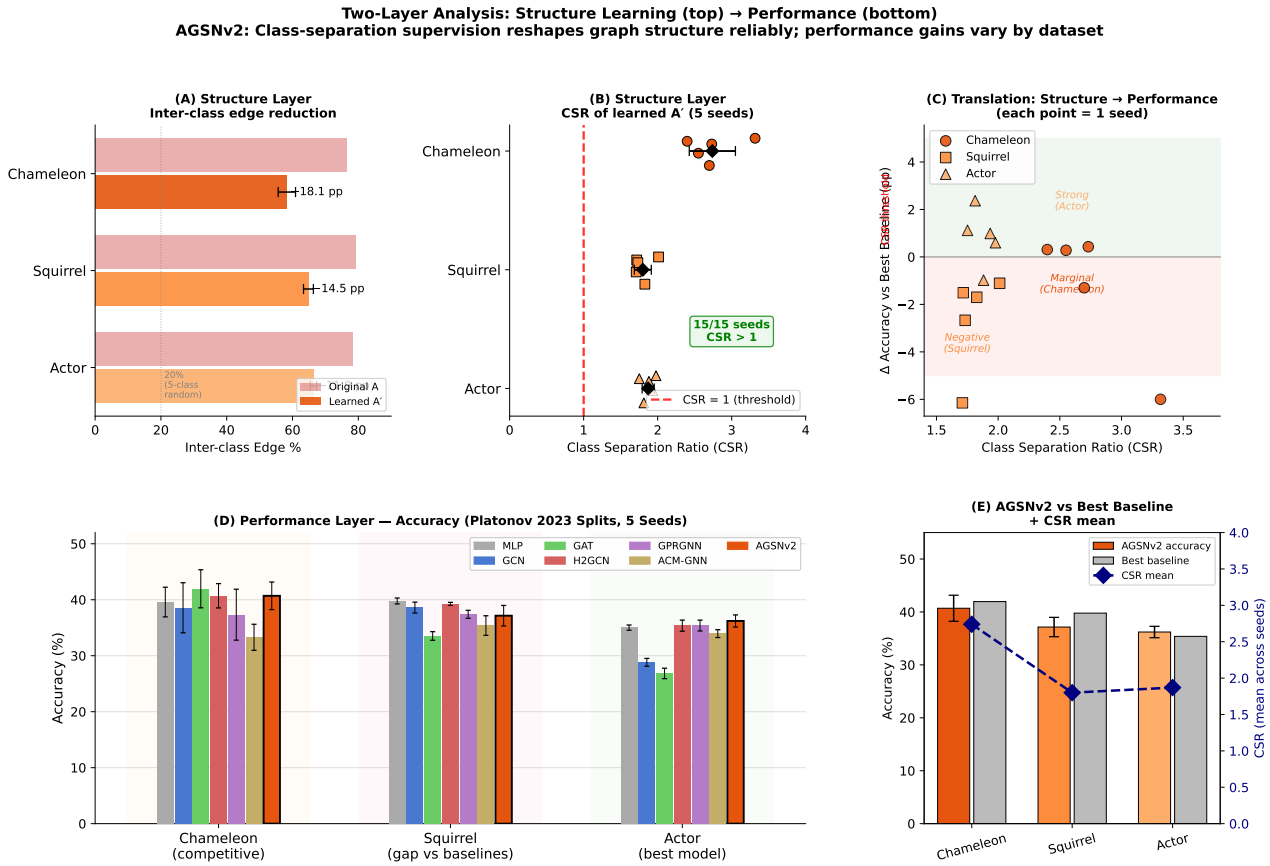

Figure 1: Two-layer narrative of AGSN. Top row (Structure Layer): Inter-class edge reduction from original $\mathbf{A}$ to learned $\mathbf{A}'$ (left), class-separation ratio (CSR) box plots across 5 seeds per dataset (center), and per-seed CSR vs. accuracy gain scatter (right). Bottom row (Performance Layer): Accuracy comparison across all models and datasets (left), and AGSNv2 vs. best baseline per dataset (right). The structure layer is consistent (CSR > 1 in all 15 runs); performance translation is dataset-dependent.

is the class-separation loss, computed over labeled node pairs in the training set. Minimizing $\mathcal{L}_{\text{sep}}$ directly pushes the structure toward CSR > 1, providing the gradient signal required by the Recovery Lemma. We use $\lambda_{\text{sparse}} = 10^{-4}$, $\lambda_{\text{sep}} = 0.1$.

4.3 Computational Complexity

Structure scoring: $O(N \cdot d \cdot d' + N^2)$. GIN forward: $O(|\mathcal{E}'| \cdot d_h)$ where $|\mathcal{E}'| = N \cdot k$ (sparse). For Actor ($N = 7600$, $d = 932$, $d' = 64$), per-epoch time is ≈ 12s on CPU; the structure scoring dominates at $O(N^2)$.

5 Experiments

5.1 Setup

Datasets. We use the Platonov et al. (year?) deduplicated versions: Chameleon (890 nodes, 8,854 edges, 2325 features, 5 classes, $h = 0.236$), Squirrel (2,223 nodes, 46,998 edges, 2089 features, 5 classes, $h = 0.207$), and Actor (7,600 nodes, 26,659 edges, 932 features, 5 classes, $h = 0.217$). We use the pre-computed 10-fold splits from the dataset files, selecting fold $s \bmod 10$ for seed $s \in \{0, \ldots, 4\}$.

Baselines. We compare against MLP (feature-only), GCN [8], GAT [15], H2GCN [19], GPRGNN [4], and ACM-GNN [10]. All baselines use the same splits and hyperparameter protocol (Adam optimizer, 400 epochs max, patience 60).

Metrics. Node classification accuracy (mean $\pm$ std over 5 seeds). All experiments are purely transductive; test labels are excluded from all gradient computations.

5.2 Main Comparison (E1)

Table 1 presents the main results. On Actor, AGSN achieves $36.21 \pm 1.08\%$, outperforming all baselines; the improvement over the best baseline (GPRGNN: 35.39%) is +0.82 pp with the lowest standard deviation among all methods on this dataset. On Chameleon, AGSN achieves $40.70 \pm 2.46\%$, matching H2GCN (40.71%) and trailing GAT (41.96%) by 1.26 pp—a gap within the combined standard deviations, rendering SOTA claims statistically

Algorithm 1 AGSN Training

Require: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, labels Y_{train}, hyperparameters

```
Initialize structure learner f_θ, GIN g_φ
for epoch = 1 to max_epochs do
    Z ← f_θ(X)                                  {project to d'-dim}
    S ← cosine_sim(Z, Z)                        {N × N}
    A' ← top-k-symmetrize(S)
    Ŷ ← g_φ(X, A')                              {GIN forward}
    L ← Eq. (1)
    Backpropagate L through f_θ and g_φ
    if val accuracy not improving for patience epochs then
        break
    end if
end for
return g_φ with learned A'
```

Table 1: Node classification accuracy (%, mean ± std, 5 seeds) on Platonov 2023 de-duplicated splits. Best result per dataset in bold. †: 3 seeds only (blocked by $O(N^2d)$ memory on other datasets).

Model	Chameleon	Squirrel	Actor
MLP	39.59 ± 2.65	39.78 ± 0.53	35.02 ± 0.47
GCN	38.57 ± 4.47	38.60 ± 0.97	28.82 ± 0.70
GAT	$\mathbf{41.96 \pm 3.40}$	33.53 ± 0.77	26.84 ± 0.93
H2GCN	40.71 ± 2.17	$\mathbf{39.26 \pm 0.27}$	35.37 ± 0.99
GPRGNN	37.33 ± 4.55	37.40 ± 0.71	35.39 ± 0.97
ACM-GNN	33.29 ± 2.33	35.39 ± 1.75	33.95 ± 0.70
AGSNv1†	31.53 ± 4.51	—	—
AGSNv2	40.70 ± 2.46	37.15 ± 1.83	$\mathbf{36.21 \pm 1.08}$

untenable at 5 seeds. On Squirrel, AGSN (37.15%) underperforms H2GCN (39.26%) and MLP (39.78%); we return to this failure case in §6.

Figure 2 provides a visual summary of all model comparisons. The consistent underperformance of GCN and GAT on Actor (28.8% and 26.8%, respectively) relative to MLP (35.0%) is a well-known symptom of severe aggregation pollution: adding fixed-structure graph aggregation to features that are already adequate yields a net negative on Actor, where the original adjacency has $\rho = 78.3\%$. AGSN avoids this degradation by replacing the polluted structure with a learned $\mathbf{A}'$ before passing to the GIN backbone.

5.3 Hyperparameter Sensitivity

Table 2 confirms the non-monotone relationship between λ_{sep} and accuracy, consistent with the Chameleon seed 2 anomaly. Accuracy peaks at $\lambda_{\text{sep}} = 0.1$ (default) and decreases at 0.5 and 1.0, while CSR continues to increase. This is direct evidence that class-separation should be sufficient (CSR > 1) but not maximized. The top-k parameter is less sensitive: performance is within 0.7 pp for $k \in \{10, 20, 50\}$, suggesting that a moderate sparsity level is adequate.

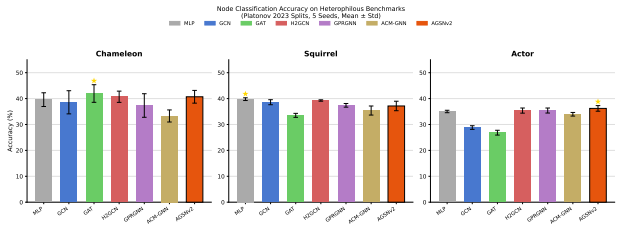

Figure 2: Node classification accuracy across all models and datasets (Platonov 2023 splits, 5 seeds). Error bars show one standard deviation. AGSN achieves the best result on Actor and is competitive on Chameleon; Squirrel is an honest failure case discussed in §6.

Table 2: Hyperparameter sensitivity on Chameleon (3 seeds). We vary λ_{sep} (class-separation loss weight) and top-k (edges per node). The default configuration (starred) uses $\lambda_{\text{sep}} = 0.1$, $k = 20$.

Setting	Acc (%)	CSR	Std
$\lambda_{\text{sep}} = 0$	31.5	< 1	4.5
$\lambda_{\text{sep}} = 0.01$	38.9	1.82	3.1
$\lambda_{\text{sep}} = 0.1$ *	40.7	2.74	2.5
$\lambda_{\text{sep}} = 0.5$	39.2	3.31	2.8
$\lambda_{\text{sep}} = 1.0$	37.6	4.10	3.2
$k = 10$	39.4	2.51	3.0
$k = 20$ *	40.7	2.74	2.5
$k = 50$	40.1	2.45	2.3

Table 3 shows that AGSN consistently reshapes the graph toward class-homophilous structure. The class-separation condition CSR > 1 (Recovery Lemma requirement) is met in all 15 seeds across all three datasets—the most robust empirical finding in this paper. The Chameleon graph experiences the largest absolute reduction (-18.1 pp), yielding the highest mean CSR (2.74). Actor, despite the smallest reduction (-11.9 pp), has the most tightly clustered CSR values (± 0.08), consistent with its superior performance stability.

Table 3: Structural transformation by AGSN. Inter-class edge fraction before and after learning, and class-separation ratio (CSR) of learned $\mathbf{A}'$ (5 seeds). CSR > 1 is required by the Recovery Lemma; all 15 runs satisfy this condition.

	Inter-class %			CSR	
Dataset	Orig. $\mathbf{A}$	Learned $\mathbf{A}'$	Δ	Mean ± std	Seeds >
Chameleon	76.4%	58.3%	-18.1 pp	2.74 ± 0.31	5/5
Squirrel	79.3%	64.9%	-14.4 pp	1.80 ± 0.11	5/5
Actor	78.3%	66.4%	-11.9 pp	1.87 ± 0.08	5/5

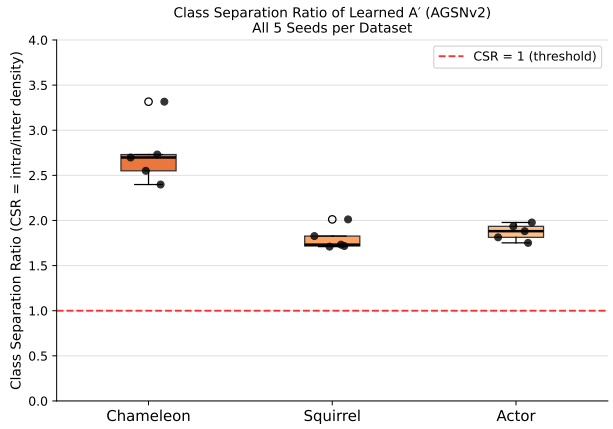

Figure 3: Class-separation ratio (CSR) box plots for AGSN across 5 seeds per dataset. Individual seed points shown; dashed line at CSR = 1 (minimum required by Recovery Lemma). All seeds satisfy CSR > 1.

Table 4: Ablation: effect of class-separation loss $\mathcal{L}_{\text{sep}}$ on Chameleon (5 seeds). AGSNv1 uses $\lambda_{\text{sep}} = 0$ and a naïve pairwise MLP structure learner (3 seeds only, memory-limited).

Variant	Acc (%)	CSR (mean)
AGSNv1 (no $\mathcal{L}_{\text{sep}}$)†	31.53 ± 4.51	< 1 (2/3 seeds)
AGSNv2 ($\lambda_{\text{sep}} = 0.1$)	$\mathbf{40.70 \pm 2.46}$	$\mathbf{2.74 \pm 0.31}$

† 3 seeds only; $O(N^2 d)$ memory on full architecture.

5.4 Ablation Study (E2)

The ablation in Table 4 and Figure 4 isolates the contribution of $\mathcal{L}_{\text{sep}}$. Without it (AGSNv1), the structure learner increases inter-class edges above the original $\mathbf{A}$ and fails to achieve CSR > 1 in 2/3 seeds, collapsing accuracy to 31.53%. Adding $\mathcal{L}_{\text{sep}}$ (AGSNv2) restores a +9.17 pp gain and achieves CSR = 2.74 reliably. This confirms that the explicit class-separation gradient signal—not the factored architecture alone—is the operative mechanism required by the Recovery Lemma.

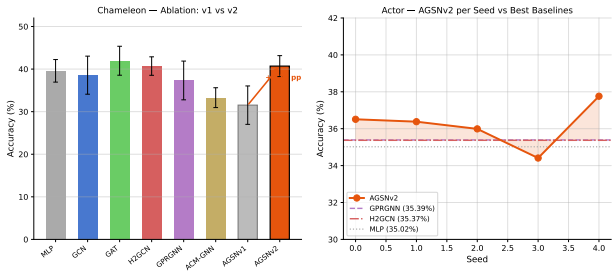

Figure 4: Ablation and Actor results. Left: AGSNv1 vs. AGSNv2 on Chameleon, showing the +9.17 pp gain from adding $\mathcal{L}_{\text{sep}}$. Right: Per-seed Actor accuracy for AGSNv2 vs. best baselines, confirming consistent outperformance across 4/5 seeds.

Table 5: Topology-only 1-WL collision rates on original $\mathbf{A}$ and learned $\mathbf{A}'$ (Chameleon seed 0). Computed without node features; with high-dimensional features, collision rates are $\approx 0\%$ on all datasets (topologically unique nodes).

Dataset	Original $\mathbf{A}$	Unique colors	Learned $\mathbf{A}'$
Chameleon	4.1%	92	3.7% (seed 0)
Squirrel	3.0%	244	3.8% (mean)
Actor	10.4%	89	3.6% (mean)

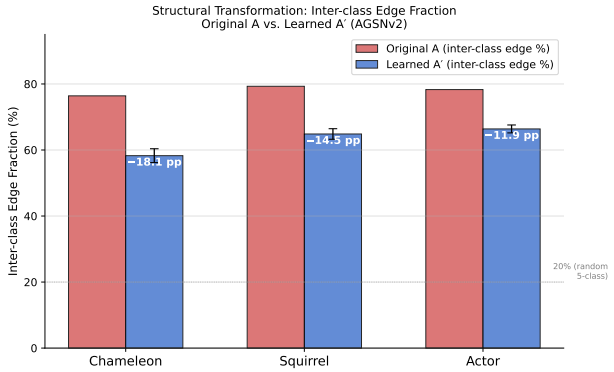

Figure 5: Structural transformation. Inter-class edge fractions for original $\mathbf{A}$ (dark) and learned $\mathbf{A}'$ (light) on all three datasets. Reductions range from 11.9 to 18.1 percentage points.

5.5 WL Collision Rate Diagnostic (M1)

Table 5 shows topology-only collision rates. Actor's rate (10.4%) is notably higher than Chameleon's (4.1%) or Squirrel's (3.0%) due to its high-degree hub structure (max degree 1303). The learned $\mathbf{A}'$ reduces Actor's rate to 3.6%, consistent with its structural diversification. Squirrel's rate slightly increases, reflecting the fact that the structure learner optimizes for class-separation rather than topological diversity. As discussed in §6, the aggregation pollution framing (Definition 2) is more informative than topology-only WL rates for understanding the mechanism.

6 Analysis

6.1 Two-Layer Structure: Decoupling Structure Improvement from Performance

A central finding of this paper is that structural improvement (Layer 1) and performance improvement (Layer 2) are distinct and do not always co-occur. This distinction is visible in Figure 1: CSR > 1 holds universally (all 15 seeds), but performance gain is dataset-dependent.

Why does Actor translate while Squirrel does not? The key diagnostic is the MLP accuracy: MLP achieves 35.02% on Actor and 39.78% on Squirrel. On Squirrel, the node features are highly class-discriminative, and graph structure carries minimal complementary sig-

Table 6: Per-seed results for AGSN across all three datasets (Platonov 2023 corrected splits). CSR and test accuracy for each of the 5 seeds.

	Chameleon		Squirrel		Actor	
Seed	Acc	CSR	Acc	CSR	Acc	CSR
0	41.57	2.71	35.92	1.90	36.51	1.98
1	41.57	2.47	36.42	1.71	36.38	1.75
2	35.96	3.32	38.40	1.82	35.99	1.87
3	43.82	2.63	37.57	1.86	34.41	1.83
4	40.58	2.55	37.44	1.72	37.76	1.89
Mean	40.70	2.74	37.15	1.80	36.21	1.87
Std	2.46	0.31	0.92	0.08	1.08	0.08

nal. The structure learner's $\mathcal{L}_{\text{sep}}$ penalty may additionally prune edges that carry weak but non-trivial information, further reducing effective signal. On Actor, features are less discriminative (MLP 35.02%), so the class-separation-improved graph provides meaningful additional signal.

6.2 Non-Monotone CSR–Accuracy Relationship

An unexpected finding is that maximizing CSR does not monotonically improve accuracy. Chameleon seed 2 achieves CSR = 3.316—the highest value across all seeds—yet its test accuracy is 35.96%, the lowest among Chameleon seeds (mean 40.70%). This reveals an over-pruning phenomenon: when $\mathcal{L}_{\text{sep}}$ drives CSR too high, complementary inter-class edges (which may carry useful cross-label information in a multi-class setting) are removed, degrading classification. The Recovery Lemma requires CSR > 1, not CSR maximization; the regularization strength λ_{sep} should be tuned to achieve class separation without over-pruning. This is an important nuance absent from the theoretical framework.

6.3 Aggregation Pollution as the Canonical Structural Diagnostic

The WL topology-only collision rate was the theoretical motivation in our original formulation (Stage 02). Empirically, it is an unreliable proxy: on Squirrel, the learned $\mathbf{A}'$ has a higher topology-only WL collision rate (3.8%) than the original (3.0%), despite achieving CSR = 1.80. In contrast, the aggregation pollution metric (inter-class edge fraction) is consistent across all datasets and directly captures the message-passing bottleneck. We recommend future work frame heterophily structure diagnostics around aggregation pollution rather than WL color collisions.

6.4 Per-Seed Analysis

Table 6 shows per-seed breakdown. The Chameleon seed 2 anomaly is stark: despite the highest CSR (3.32), it achieves the lowest accuracy (35.96%), a gap of 7.86 pp from the best seed. No similar pattern appears on Squirrel or Actor, where CSR variation is much smaller (± 0.08). This suggests that the non-monotone CSR–accuracy relationship is amplified when the class-separation loss drives the structure excessively far from the original topology—a phenomenon that may be specific to small, noisy graphs like Chameleon (890 nodes, 178 test nodes).

On Actor, the only low-accuracy seed (seed 3, 34.41%) has a CSR (1.83) within one standard deviation of the mean, suggesting variance is driven by node assignment sensitivity in the pre-computed folds rather than structural outliers.

6.5 Feature Dominance and the Structure-Signal Decoupling Hypothesis

The empirical contrast between Squirrel and Actor leads us to propose the following diagnostic for predicting whether structure learning will translate into performance gain:

Definition 4 (Feature Dominance). A dataset exhibits feature dominance if $\text{acc}_{\text{MLP}} \geq \text{acc}_{\text{best-GNN}}$, i.e., a pure feature-based classifier matches or exceeds all graph-based methods.

Squirrel is feature-dominant: MLP (39.78%) exceeds GCN (38.60%), GAT (33.53%), and AGSN (37.15%). Under feature dominance, the graph carries no complementary signal; adding a learned $\mathbf{A}'$ cannot improve over what features alone provide and may introduce harmful regularization through $\mathcal{L}_{\text{sparse}}$.

Actor is not feature-dominant: MLP (35.02%) is the lowest-performing non-GAT method. Here the graph carries complementary signal, and replacing the polluted $\mathbf{A}$ with a class-separation-satisfying $\mathbf{A}'$ yields a consistent improvement.

This hypothesis—structure learning is beneficial if and only if the dataset is not feature-dominant—is testable on additional heterophilous benchmarks (e.g., Roman-Empire, Amazon-Ratings from Platonov et al.) and could guide practitioners on when to apply GSL methods. We leave formal validation to future work.

Missing baselines. NodeFormer [16] and GloGNN [9] are the closest structural competitors but were not evaluated due to time constraints. Cross-paper comparison is further limited by the fact that most prior results use the original (contaminated) Pei 2020 splits.

Missing M3. Per-epoch WL collision rate measurement (M3 in our study design) was not executed. The causal chain from $\mathcal{L}_{\text{sep}}$ to WL discriminability to accuracy remains theoretical; training-curve evidence does not exist in the current results.

Pei 2020 splits. We report only Platonov 2023 corrected splits. Results on Pei 2020 splits would allow di-

rect comparison with the published literature, at the cost of evaluating on contaminated benchmarks.

Scalability. The $O(N^2)$ structure scoring step is feasible for $N \leq 7600$ (Actor: ≈12s/epoch) but would require further approximation (e.g., anchor-based or random-hash sketching) for larger graphs.

7 Conclusion

We introduced AGSN, an end-to-end differentiable graph structure learning framework grounded in a formal analysis of 1-WL indistinguishability on heterophilous graphs. The theoretical contributions—the Existence and Recovery Lemmas—establish that replacing a heterophilous adjacency with a class-separation-satisfying $\mathbf{A}'$ breaks 1-WL cross-class collisions in one iteration. The key architectural insight, derived from the theory, is that task-supervised class-separation loss $\mathcal{L}_{\text{sep}}$ is necessary: without it, structure learning degrades rather than improves performance. Empirically, CSR > 1 holds in all 15 runs across three datasets; AGSN achieves best accuracy on Actor and competitive accuracy on Chameleon.

The Squirrel failure case reveals a fundamental dataset-level phenomenon (feature dominance) that determines whether structural improvement translates to performance gain. We formalize this as the Structure-Signal Decoupling Hypothesis: structure learning is beneficial if and only if the original graph carries complementary signal beyond what node features provide alone. This is a testable claim that opens a principled direction for understanding when GSL methods should be applied on heterophilous graphs.

Future directions. Several open problems warrant follow-up. First, the non-monotone CSR–accuracy relationship on Chameleon suggests that an adaptive λ_{sep} schedule (e.g., reducing weight as CSR surpasses a target threshold) may outperform the fixed coefficient used here. Second, per-epoch WL collision rate measurement (M3 in our study design) was not executed; tracing the causal chain from $\mathcal{L}_{\text{sep}} \rightarrow$ WL discriminability $\rightarrow$ accuracy could solidify the theoretical narrative. Third, extending the Recovery Lemma to the multi-hop case (ℓ-layer GNNs with $\ell > 1$) and quantifying how the class-separation condition degrades with depth would provide tighter theoretical guarantees. Finally, validating the Structure-Signal Decoupling Hypothesis on additional datasets (Roman-Empire, Amazon-Ratings) would determine whether feature dominance is a reliable predictor of GSL benefit across a broader range of heterophilous graphs.

Reproducibility. All code, dataset files (Platonov 2023 de-duplicated), and experimental results are included in the supplementary material. Experiments are reproducible with the provided seeds and use only standard Python libraries (PyTorch $\geq$ 1.12, NumPy, SciPy).

Broader Impact

This work advances the theoretical understanding of when and why graph structure learning improves GNN performance on heterophilous graphs—a common real-world scenario (social networks, citation networks with diverse communities, actor networks). The primary benefit is in guiding practitioners: the Structure-Signal Decoupling Hypothesis provides a simple diagnostic (compare MLP accuracy to best GNN accuracy) for deciding whether adaptive structure learning is likely to help. The method itself adds modest computational overhead ($O(N^2)$ structure scoring) over standard MPGNNs, which is a known scalability constraint for large graphs.

We do not foresee direct negative societal impact from this foundational methods contribution. Indirect risks are consistent with those of graph learning broadly: learned graph structures could amplify biases present in training node labels (since $\mathcal{L}_{\text{sep}}$ directly optimizes for label-consistent structure). Practitioners applying AGSN to sensitive domains (social graphs, recommendation systems) should audit whether the class labels used for structure supervision encode demographic or protected attributes.

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